

Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

Asaf A. Solonnikov and Nadav Katz

The Racah institute of physics, The Hebrew University of Jerusalem, Israel

(Dated: June 13, 2026)

We present a qubit spectroscopy method based on Lorentzian-derived drive pulses that enables T_2 -limited frequency characterization at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude, thereby enhancing robustness against amplitude variations. Specifically, we employ a root-Lorentzian envelope with a π -phase inversion at its peak, which introduces an echo-like symmetry that cancels on-resonant Rabi oscillations while preserving the spectral response. This approach suppresses conventional power broadening and enables high-resolution spectroscopy using pulses shorter than durations required by standard techniques. We experimentally demonstrate this method on a superconducting qubit platform and observe excellent agreement with numerical simulations, highlighting its efficiency and reliability for rapid and calibration-robust qubit frequency characterization.

Accurate frequency characterization of a quantum two-level system (qubit) is essential for calibrating high-fidelity control in fault-tolerant quantum processors. In many physical implementations, such as superconducting circuits and semiconductor quantum dots, the qubit is realized within an inherently multilevel structure with discrete energy levels, allowing two levels to be selected to form an effective computational subspace that mimics atomic systems via their quantized energy spectrum [1, 2]. Spectroscopic identification of the transition frequency is therefore routinely performed by applying resonant control fields and monitoring the resulting population dynamics [3–5]. Spectroscopic resolution in coherently driven two-level systems is fundamentally constrained by decoherence. In the absence of technical noise, stochastic phase fluctuations induce homogeneous broadening of the transition frequency, leading to a minimum resolvable linewidth that scales inversely with the coherence time T_2 , as described by the Bloch equations [6, 7]. This establishes the so-called T_2 limit, which is generally regarded as a fundamental bound on the spectral distinguishability of energy-level transitions.

In practice, however, driven spectroscopy introduces an additional trade-off between signal robustness and spectral resolution. When finite-amplitude control pulses are applied, the system dynamics deviate from the idealized two-level response due to coherent Rabi oscillations and coupling to noncomputational levels. Increasing the drive amplitude improves signal contrast and robustness against state-preparation and measurement errors, but simultaneously induces power broadening, resulting in a linewidth that scales with the instantaneous Rabi frequency. Consequently, conventional spectroscopic techniques must operate at low drive amplitudes in order to maintain resolution approaching the decoherence-limited linewidth set by T_2 , rendering precision and robustness fundamentally incompatible.

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit. Achieving such resolution, however, requires long interrogation times and sufficiently dense temporal sampling to resolve the re-

sulting interference fringes. Moreover, Ramsey measurements are susceptible to systematic biases: asymmetries in the exponential decay envelope—arising, for example, from pulse miscalibration—can shift the apparent resonance peak in the frequency domain. Additional effects such as phase fluctuations or unintended AC Stark shifts during the control pulses introduce further distortions, ultimately biasing the extracted transition frequency.

Continuous efforts aim to mitigate this trade-off through dynamical decoupling or engineered dissipation [8–10]. However, these approaches typically require extended interrogation times or additional control resources, and do not overcome the intrinsic connection between decoherence and spectral resolution, as the same stochastic frequency fluctuations that govern dephasing also determine the resolvability of the transition frequency. Here, we show that this limitation can be bypassed in the driven regime by employing adiabatically engineered Lorentzian-derived drive pulses with a π -phase inversion at their peak. This echo-like structure suppresses on-resonant Rabi oscillations while preserving the steady-state spectral response, enabling T_2 -limited frequency characterization at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude.

Precise determination of the qubit transition frequency is crucial for implementing high-fidelity quantum gates. The drive frequency must coincide with the qubit resonance, as detuning directly reduces gate fidelity. A rigorous metric for quantifying such calibration errors is the diamond norm, which captures deviations in the implemented quantum operation from the ideal one. Minimizing these detuning-induced errors is essential to meet the fault-tolerance thresholds required for scalable quantum computation. [11]

Here, we propose a spectroscopic method for probing the qubit transition frequency using arbitrarily shaped drive pulses, specifically an echo-root-Lorentzian pulse. In previous works by Vitanov [12], it was shown that Lorentzian pulses exhibit power-narrowing, complementary to the conventional power broadening observed for constant drives, originating from the adiabatic proper-

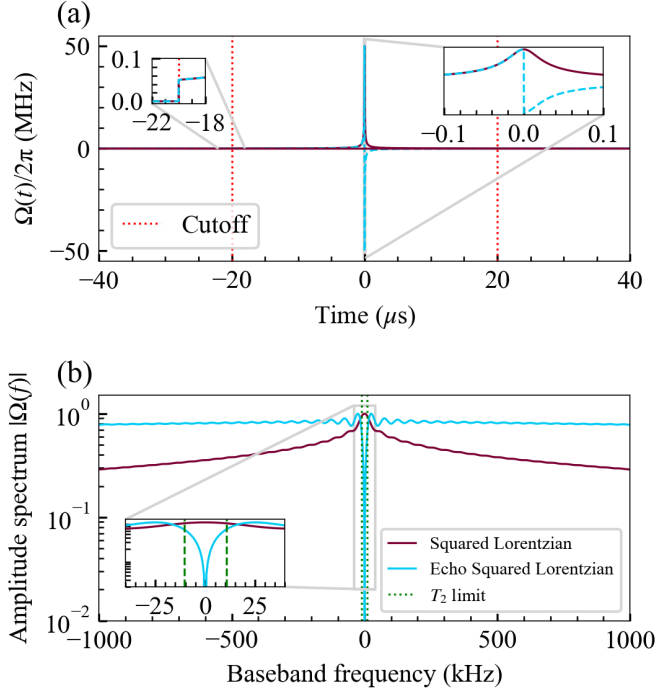


Figure 1. Time and frequency-domain representation of the root-Lorentzian and echo-root-Lorentzian drive pulses. (a) Time-domain profiles of a Lorentzian $L_{1/2}$, and echo-Lorentzian $L_{1/2}^{\text{echo}}$ pulse with a π phase jump at $t = 0$. Red dotted lines indicate the temporal cutoff applied to the pulse envelope. Insets show zoomed-in views of the pulse near the cutoff and phase-jump regions. (b) Corresponding amplitude spectra $|\Omega(f)|$. The echo-root-Lorentzian pulse exhibits a suppression of spectral weight around zero frequency due to destructive interference induced by the phase inversion, resulting in a spectral null at resonance. The T_2 -limited linewidth is indicated by the green dotted line.

ties of these pulses. In this work, we extend the analysis of Vitinov to longer pulse durations approaching the T_1 timescale, thereby entering a regime beyond the Fourier limit. The echo property arises from a π -phase jump applied at the peak amplitude of the pulse, which effectively cancels its net area, while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the root-echo-Lorentzian $L_{\text{echo}}^{(1/2)}$ drive envelope based on ordered Lorentzian $L^{(1/2)}$ as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \quad (1)$$

$$L_{\text{echo}}^{(1/2)}(t) := L^{(1/2)}(t) \text{sgn}(t). \quad (2)$$

where τ sets the characteristic pulse duration and n controls the temporal sharpness.

Under the adiabatic condition for the plain Lorentzian

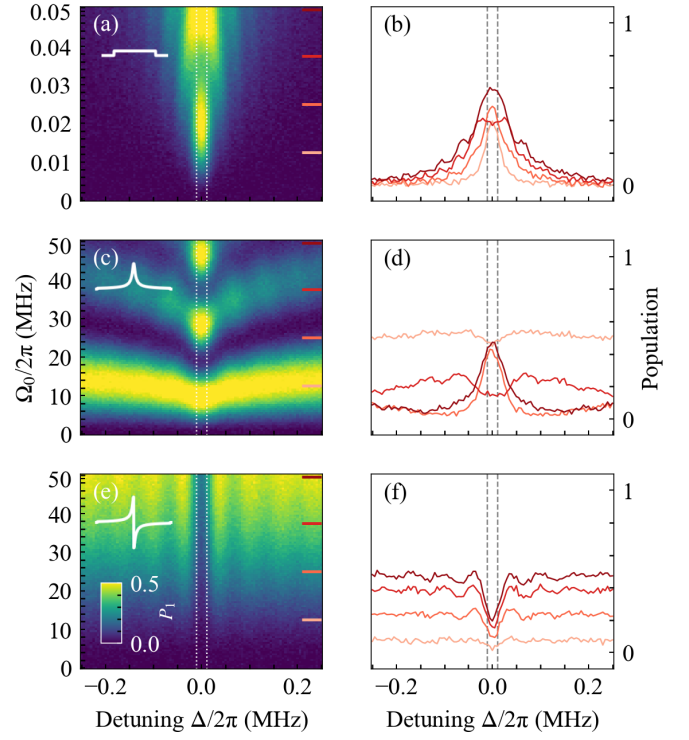


Figure 2. Two-dimensional spectroscopy of the excited-state population as a function of detuning Δ and drive amplitude Ω_0 for three pulse types: (a) square, (c) root-Lorentzian, and (e) echo-root-Lorentzian pulses. Panels (b), (d), and (f) show detuning line cuts at selected amplitudes $\Omega_0/2\pi$, as indicated in the corresponding colored ticks on the left column plots. Vertical dashed lines denote the T_2 limit line width. The square pulse exhibits strong power broadening already at small amplitudes ($\Omega_0/2\pi \sim 0.05$ MHz), whereas the root-Lorentzian and echo-type pulse (30 μs) maintain narrow, T_2 -limited linewidths. The echo-type pulse further suppresses coherent vertical oscillations, enabling smooth high-resolution spectroscopy all over the amplitude range, but with an expense of SNR.

pulse, the characteristic spectral width obeys the scaling

$$\Delta_{1/2}\tau \propto (\Omega_0\tau)^{-\frac{1}{2n-1}}, \quad (3)$$

where $\Delta_{1/2}$ denotes the typical linewidth and Ω_0 is the peak Rabi frequency. This relation indicates that increasing the drive amplitude suppresses the detuning at which adiabaticity breaks down, resulting in a narrowing of the effective spectral response. In contrast, the square pulse is fully governed by conventional power broadening, for which the linewidth increases approximately linearly with the drive amplitude $\Delta_{1/2} \propto \Omega_0$. This contrast highlights that the spectroscopic response is not determined solely by the drive strength, but depends sensitively on the temporal envelope of the control pulse, motivating the use of tailored pulse shapes [13–17].

Spectroscopy results obtained using the echo-root-Lorentzian pulse, together with standard square and root-Lorentzian pulses for comparison, are shown in Fig.

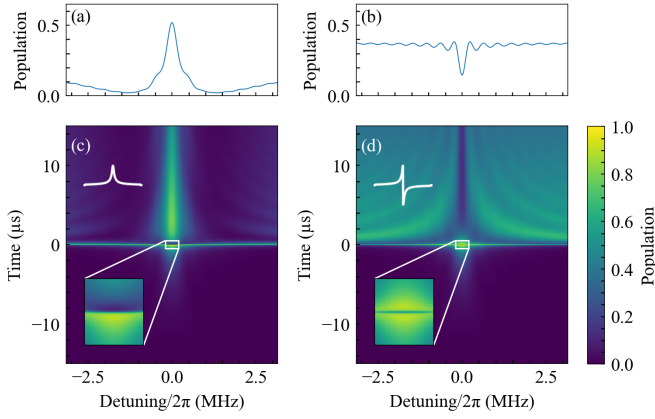


Figure 3. Numerical simulation of qubit spectroscopy under a root-Lorentzian drive and its echo variant. (a) Final excited-state population as a function of detuning $\Delta/2\pi$ for a standard root-Lorentzian pulse. (b) Same for the echo-root-Lorentzian pulse $L_{\text{echo}}^{(1/2)}$. (c),(d) Full time evolution of the excited-state population versus detuning for the corresponding drives. The pulse duration is $30 \mu\text{s}$ with a cutoff of 5×10^{-4} . While the standard pulse yields a resonant peak at zero detuning, the echo sequence produces an inversion of the central response, leading to a pronounced dip. This behavior reflects cancellation of coherent oscillations at $\Delta = 0$, as highlighted in the insets.

2. For the measured coherence time $T_2 \approx 14 \mu\text{s}$, the corresponding coherence-limited linewidth is $\Delta_{1/2}/2\pi \approx 20 \text{ kHz}$. This limit is indicated by a dotted white line in the two-dimensional amplitude-detuning sweep and by gray dashed lines in one-dimensional spectroscopy slices extracted at constant drive amplitude.

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior, enabling resolution approaching the T_2 -limited linewidth. However, for pulse durations comparable to the qubit lifetime T_1 , coherent oscillations arise for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase, such that the corresponding forward and backward evolutions cancel exactly at zero detuning. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. 1) In the frequency domain, this manifests as a splitting of the spectral response accompanied by a null at resonance, rendering the measured linewidth largely insensitive to drive amplitude. Formally, this mechanism is equivalent to a Loschmidt-echo-type evolution, in which reversibility is achieved only on resonance, while detuning

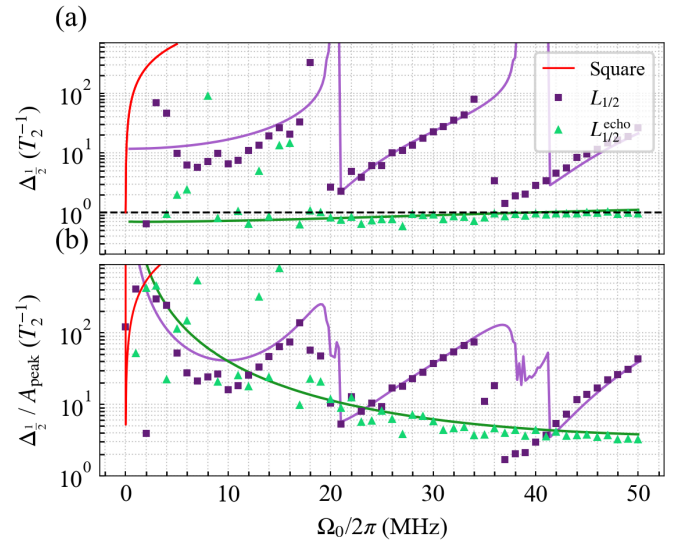


Figure 4. Comparison of spectroscopic linewidth obtained using a constant (square), Lorentzian $L^{(1/2)}$, and echo-Lorentzian $L_{\text{echo}}^{(1/2)}$ drive as a function of peak Rabi frequency Ω_0 . (a) Measured full width at half maximum (FWHM) $\Delta_{1/2}$ of the qubit spectral response, normalized to the T_2^{-1} limit, together with numerical simulations. Solid lines denote simulation results, while markers represent experimental data. The constant drive (red) exhibits conventional power broadening $\Delta_{1/2} \propto \Omega_0$, whereas the Lorentzian (green) and echo-Lorentzian (purple) pulses display power narrowing due to adiabatic following. (b) Linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$, highlighting the improved resolution-signal trade-off enabled by the echo-Lorentzian protocol at increasing drive amplitude.

or decoherence breaks the cancellation and restores excitation. Combined with the adiabatic following ensured by the Lorentzian envelope, this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling spectroscopy approaching the T_2 -limited linewidth over a broad range of drive strengths and with pulse durations substantially shorter than those required by standard spectroscopy or Ramsey interferometry.

Fig. 3 shows a numerical simulation of the qubit state evolution during the spectroscopy echo-root-Lorentzian sequence. At resonance, where the adiabatic condition breaks down, coherent Rabi oscillations are induced by the drive. However, due to the π phase inversion at the pulse maximum, the subsequent segment of the drive reverses the accumulated evolution, resulting in a symmetric cancellation that returns the qubit close to the ground state, limited by relaxation ($T_1 = 30 \mu\text{s}$).

For detuned drives, the system remains in the adiabatic regime throughout most of the pulse duration and undergoes excitation only in the vicinity of the pulse peak, where the phase inversion occurs. In this case, the evolution does not cancel and results in a net population transfer followed by relaxation. As a consequence, different detunings produce distinguishable final states, forming the basis of the spectroscopic response.

In Fig. 5, we plot the extracted spectroscopic linewidth obtained from Gaussian fits to the measured response. Although the spectral lineshape of a driven two-level system is often expected to follow a Lorentzian profile, the interference-based structure of the present protocol results in a more complex response. In this regime, Gaussian fitting provides a robust and consistent estimate of the linewidth while yielding values comparable to those obtained from Lorentzian fits within our resolution requirements.

We compare the experimentally acquired linewidth extracted from two-dimensional frequency–amplitude sweeps with numerical simulations of an effective two-level system using identical system parameters. The square (red), Lorentzian (green), and echo-Lorentzian (purple) drives are shown, where solid lines denote simulation results and markers represent experimental data. For the square pulse, only simulated results are displayed, as the agreement with experiment was independently validated. We plot both the linewidth $\Delta_{1/2}$ and the linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$ to quantify the resolution–signal trade-off.

To suppress residual power broadening at large drive amplitudes the pulse envelope must satisfy requirements on its cutoff behavior. The cutoff effectively introduces a discontinuity in the envelope derivative, effectively behave as a constant pulse, producing a linear broadening that scales with the instantaneous amplitude at the cut-off point, denoted Ω_c . For T_2 limited spectroscopy the T_2 limit $T_2^{-1} \leq \Delta_{1/2}$ therefore yields a constraint on the maximum allowable cutoff amplitude $\Omega_c < T_2^{-1}$. In our system Satisfying the above inequality requires a relative cutoff level on the order of 10^{-4} to 10^{-5} depending on pulse duration and peak drive amplitude. Empirically, we find that a cutoff less than 10^{-3} is sufficient to eliminate excess broadening and to maintain spectroscopy in the T_2 limited regime even under strong driving conditions. See Supplemental material [999].

At the core of assessing pulse quality and identifying optimal operating regimes lies the spectral response of the driven qubit. We find that the echo pulse yields spectroscopic features with markedly reduced linewidths over a well-defined region of parameter space. In particular, the pulse duration must remain shorter than the qubit lifetime T_1 , while exceeding approximately $10 \mu\text{s}$ in order to suppress Fourier-limited broadening (see Supplemental Material [999]).

This constitutes an additional advantage of the echo protocol. While both constant and simple Lorentzian pulses exhibit coherent Rabi oscillations that persist even for pulse durations comparable to or exceeding T_1 , the intrinsic π -phase inversion of the echo pulse suppresses these coherence-induced oscillations. As a result, the spectral response remains smooth and well-behaved, enabling reliable linewidth extraction in regimes where standard and Lorentzian pulses display pronounced oscillatory distortions. (see Supplemental Material [999]).

Second, the use of formally infinite-support pulses ne-

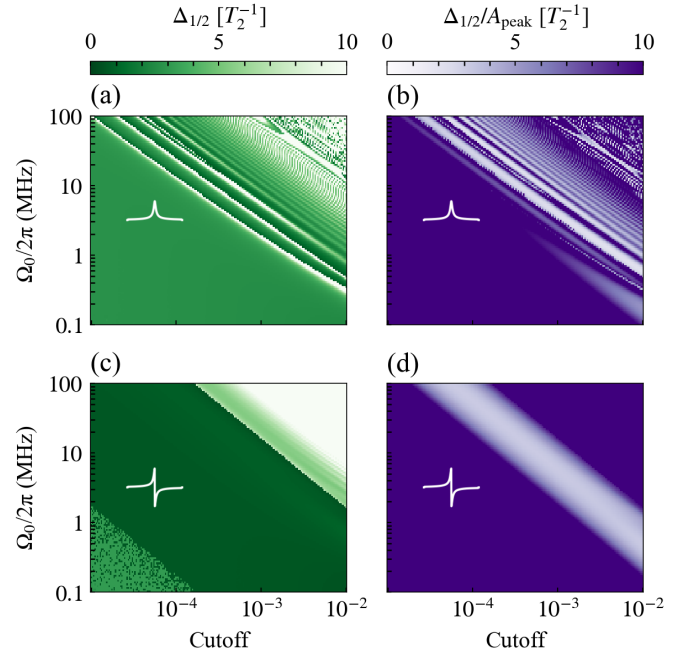


Figure 5. Numerical simulation of spectroscopy scans using root-Lorentzian $L^{1/2}$ and echo-root-Lorentzian $L_{\text{echo}}^{(1/2)}$ drives with a pulse duration of $50 \mu\text{s}$. (a),(b) Line width $\Delta_{1/2}$ and the linewidth-to-signal ratio $\Delta_{1/2}/A_{\text{peak}}$ obtained using the root-Lorentzian pulse $L^{(1/2)}$ as a function of peak Rabi frequency Ω_0 and temporal cutoff. (c),(d) Corresponding results for the echo-root-Lorentzian pulse $L_{\text{echo}}^{(1/2)}$. While $L^{(1/2)}$ pulse exhibits extended regions of narrow linewidth, the associated linewidth-to-signal ratio displays strong oscillatory behavior, making it difficult to identify robust operating parameters for high-resolution spectroscopy. In contrast, the echo variant $L_{\text{echo}}^{(1/2)}$ yields both reduced linewidth and a broad parameter region in which the resolution–signal trade-off remains favorable, enabling stable narrowband spectroscopy.

cessitates the introduction of a cutoff parameter, as discussed previously. Consequently, the relevant parameter space is effectively three-dimensional: pulse duration, drive amplitude, and cutoff level.

Motivated by the stability observed at fixed pulse duration, we restrict the analysis to constant-length pulses and examine the remaining two parameters. In Fig. 4 we simulate both the root-Lorentzian pulse and its echo-modified counterpart. We extract the full width at half maximum (FWHM) via fitting (as in Fig. 3) and evaluate the normalized metric given by the linewidth divided by the peak signal amplitude. This quantity is strongly correlated with spectral resolution and is directly connected to the inverse Fisher information, i.e., the Cramér–Rao bound.

The results reveal that the root Lorentzian pulse achieves narrow linewidths across a relatively large region of parameter space. However, its performance is not uniform and exhibits oscillatory structure for the chosen pulse duration. In contrast, the echo-root Lorentzian

pulse produces even narrower linewidths while maintaining a significantly broader and more uniform high-resolution region. Although the optimal region may appear small at first glance, the logarithmic scale of the parameter axes should be noted. More importantly, the high-resolution region forms a robust and contiguous “island,” indicating strong parameter insensitivity. This robustness makes the echo protocol substantially more stable against parameter variations and therefore highly advantageous for practical spectroscopy.

In conclusion, we have demonstrated that the echo–Lorentzian pulse constitutes a promising and practical approach for high-resolution spectroscopy. By com-

binning adiabatic shaping with an intrinsic echo phase inversion, the protocol significantly enhances robustness against drive-amplitude variations. Remarkably, it broadens the operational amplitude window by approximately three orders of magnitude, while maintaining high spectral resolution even at drive strengths approaching neighboring qubit transitions. These properties make the echo–Lorentzian protocol particularly attractive for precision qubit characterization under realistic experimental constraints.

This opens the route to coherent spectroscopy based on phase-sensitive off-resonant response rather than population transfer.

-
- [1] A. Blais, R.-S. Huang, A. Wallraff, S. M. Girvin, and R. J. Schoelkopf, *Phys. Rev. A* **69**, 062320 (2004).
 - [2] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, *Phys. Rev. A* **76**, 042319 (2007).
 - [3] H. Paik, D. I. Schuster, L. S. Bishop, G. Kirchmair, G. Catelani, A. P. Sears, B. R. Johnson, M. J. Reagor, L. Frunzio, L. I. Glazman, S. M. Girvin, M. H. Devoret, and R. J. Schoelkopf, *Phys. Rev. Lett.* **107**, 240501 (2011).
 - [4] L. B. Nguyen, Y.-H. Lin, A. Somoroff, R. Mencia, N. Grabon, and V. E. Manucharyan, *Phys. Rev. X* **9**, 041041 (2019).
 - [5] S. Kim, H. Terai, T. Yamashita, W. Qiu, T. Fuse, F. Yoshihara, S. Ashhab, K. Inomata, and K. Semba, *Communications Materials* **2**, 98 (2021).
 - [6] F. Bloch, *Phys. Rev.* **70**, 460 (1946).
 - [7] S. Zienau, *Physics Bulletin* **26**, 545 (1975).
 - [8] A. Rotem, T. Gefen, S. Oviedo-Casado, J. Prior, S. Schmitt, Y. Burak, L. McGuinness, F. Jelezko, and A. Retzker, *Phys. Rev. Lett.* **122**, 060503 (2019).
 - [9] T. Gefen, A. Rotem, and A. Retzker, *Nature Communications* **10**, 4992 (2019).
 - [10] Q. Cao, G. T. Genov, Y. Chu, J. Cai, Y. Liu, A. Retzker, and F. Jelezko, *Physical Review Letters* **135** (2025), 10.1103/llfg-s4rr.
 - [11] R. Blume-Kohout, J. K. Gamble, E. Nielsen, K. Rudinger, J. Mizrahi, K. Fortier, and P. Maunz, *Nature Communications* **8**, 14485 (2017).
 - [12] I. I. Boradjiev and N. V. Vitanov, *Optics Communications* **288**, 91 (2013).
 - [13] N. Vitanov, B. Shore, L. Yatsenko, K. Böhmer, T. Halfmann, T. Ricles, and K. Bergmann, *Optics Communications* **199**, 117 (2001).
 - [14] T. Halfmann, T. Ricles, N. Vitanov, and K. Bergmann, *Optics Communications* **220**, 353 (2003).
 - [15] G. S. Vasilev and N. V. Vitanov, *Phys. Rev. A* **70**, 053407 (2004).
 - [16] P. R. Berman, L. Yan, K.-H. Chiam, and R. Sung, *Phys. Rev. A* **57**, 79 (1998).
 - [17] N. Rosen and C. Zener, *Phys. Rev.* **40**, 502 (1932).

I. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented using a Quantum Machines OPX platform together with microwave components from Keysight Technologies, including IQ mixers, local oscillators (LOs), attenuators, filters, and amplifiers.

A schematic of the setup is shown in Fig. 6. At room temperature, baseband I/Q waveforms are generated by arbitrary waveform generators (AWGs) and upconverted to microwave frequencies using IQ mixers driven by LOs. Separate lines are used for qubit drive and resonator readout. In addition, a dedicated DC bias line provides static flux control.

The microwave signals are routed into the dilution refrigerator through heavily attenuated and filtered coaxial lines. Attenuators are distributed across multiple temperature stages (300 K to ~ 9 mK) to thermalize the lines and suppress room-temperature noise. Additional filtering (low-pass and infrared filters) further reduces high-frequency noise before the signals reach the device.

The quantum device is mounted at the mixing chamber stage (~ 9 mK). The outgoing readout signal is routed through a chain of circulators that isolate the qubit from amplifier back-action, followed by cryogenic amplification using a high-electron-mobility transistor (HEMT) amplifier at the 4 K stage. The signal is then further amplified, downconverted, and digitized at room temperature.

Digitized signals are processed in real time and used for classical post-processing and iterative experiment execution.

II. T_2 LIMIT

The Bloch equations [6] provide a theoretical framework for describing the dynamics of a two-level system under an external excitation field with strength Ω_0 , frequency detuned from natural qubit frequency with detuning Δ , including presence of relaxation and dephasing times T_1 and T_2 . Originally it was developed in the context of NMR, these equations are equally applicable to an superconducting qubits.

The formalism is based on solving a system of differential equations for the expectation values of the qubit's Pauli operators, σ_x , σ_y and σ_z , which together give full information of the qubit state.

A key advantage of the Bloch solution is its validity across a broad range of excitation pulse durations, from short to long, and for both weak and strong driving fields (however, to prevent leakage into higher energy levels, the drive amplitude must remain below the qubit's anharmonicity).

In our study, we focus on the regime where the solution to the Bloch equations describes long-duration pulses, such that the qubit undergoes complete energy saturation

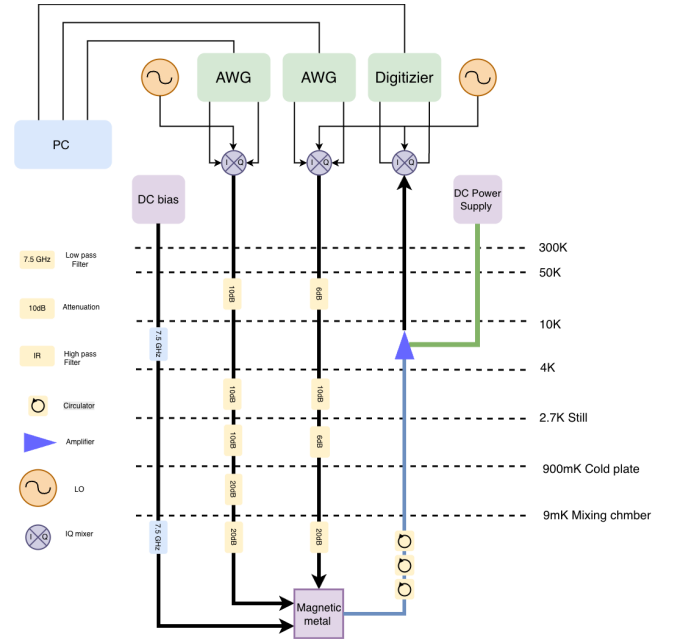


Figure 6. Schematic of the experimental setup for qubit control and readout. Room-temperature electronics (top) include a computer controlling two arbitrary waveform generators (AWGs) and a digitizer. The AWGs generate baseband I/Q signals that are upconverted to microwave frequencies via IQ mixers driven by local oscillators (LOs), enabling qubit drive and resonator readout. The signals are routed into a dilution refrigerator (dashed region), where multiple temperature stages (38 K, 10 K, 2.7 K, 900 mK, and 9 mK) are indicated. Input lines are attenuated and filtered at each stage to suppress thermal noise before reaching the device. The output signal is routed through circulators and amplified by a cryogenic high-electron-mobility transistor (HEMT) amplifier before being further processed at room temperature and digitized. The quantum device is mounted at the base temperature stage (9 mK).

$t \gg T_1$. In this limit, for resonance pulses, the qubit reaches a mixed state, and its evolution is a simple closed-form solution. In particular, we are interested in the expectation value $\langle \sigma_z \rangle$, which serves as the measurable observable of our system:

$$\langle \sigma_z \rangle = \frac{1 + (T_2 \Delta)^2}{1 + (T_2 \Delta)^2 + \Omega^2 T_1 T_2}. \quad (4)$$

The solution determines the linewidth at broad range of cases and will be applicable to understand the limits for high resolution spectroscopy.

A. Linewidths and saturation spectroscopy

When the Rabi frequency is dominant, i.e., $\Omega_0 \gg \sqrt{\frac{T_2}{T_1}}\Delta$, Eq. (4) simplifies to:

$$\langle\sigma_z\rangle \approx 1 - 2 \frac{1}{1 + \frac{\Delta^2}{T_1/T_2\Omega_0^2}}. \quad (5)$$

In this regime, the linewidth broadens as:

$$\Delta_{1/2} = \frac{2\Omega_0}{\sqrt{T_2/T_1}} \quad (6)$$

This broadening, similar to what we observe in the Rabi formula but now including saturation effects, arises due to population inversion.

Conversely, in the weak excitation regime, where $\Omega \ll \sqrt{\frac{T_2}{T_1}}\Delta$, the qubit transition linewidth is determined solely by pure dephasing T_2 . Eq. (4) reduces to:

$$\langle\sigma_z\rangle \approx 1 - 2\Omega^2 T_1 T_2 \frac{1}{1 + T_2^2 \Delta^2}, \quad (7)$$

which determines the minimal achievable linewidth—the T_2 limit:

$$\Delta_{1/2, T_2} = \frac{2}{T_2}. \quad (8)$$

III. APPENDIX CUTOFF

The spectral linewidths corresponding to the two regimes are given by

$$\delta f_1 = \frac{1}{\pi T_2}, \quad \delta f_2 = \frac{\Omega_c}{\pi \sqrt{T_2/T_1}}.$$

Requiring $\delta f_2 < \delta f_1$ yields

$$\Omega_c < \frac{1}{\sqrt{T_1 T_2}}.$$

If we express the coupling strength as $\Omega_c = c\Omega_0$, the condition becomes

$$c < \frac{1}{\Omega_0 \sqrt{T_1 T_2}}.$$

Substituting $T_1 = 30 \mu\text{s}$, $T_2 = 15 \mu\text{s}$, and $\Omega_0 = 50 \text{ MHz}$ gives

$$c < 9.4 \times 10^{-4},$$

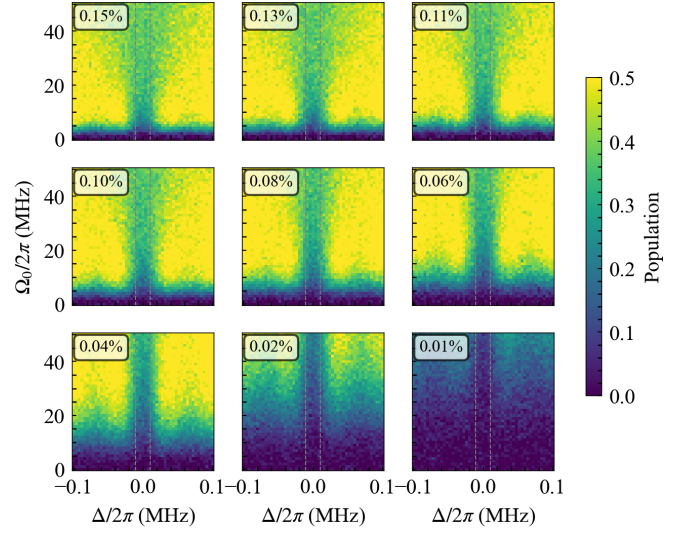


Figure 7. Spectroscopic response obtained using echo-root-Lorentzian pulses for different relative cutoff amplitudes (indicated in the upper-left corner of each panel). Increasing the cutoff smooths the phase inversion point of the pulse, thereby suppressing derivative discontinuities that otherwise induce residual power broadening. As a result, the spectral line becomes progressively narrower and more robust to drive-amplitude variations. However, for excessively small cutoff values the effective signal-to-noise ratio deteriorates, limiting the attainable spectral contrast despite reduced broadening.

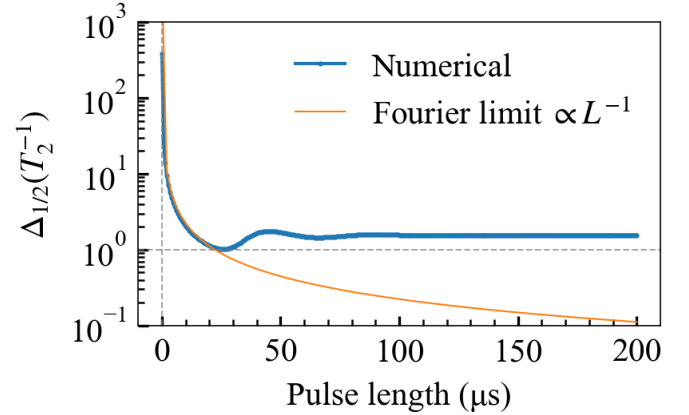


Figure 8. Numerical linewidth $\Delta_{1/2}$, normalized to the decoherence-limited scale T_2^{-1} , obtained from square-pulse spectroscopy as a function of pulse duration. For short interrogation times, the linewidth decreases inversely with pulse length, following the Fourier limit. For longer pulses, the linewidth saturates at the fundamental T_2 limit imposed by decoherence, deviating from the Fourier-limited scaling.

IV. ADIABATIC BASIS

The matrix form of the Hamiltonian in the rotated frame takes the form:

$$H(t) = \frac{1}{2} \begin{bmatrix} -\Delta & \Omega(t) \\ \Omega(t) & \Delta \end{bmatrix}. \quad (9)$$

The Rabi frequency $\Omega(t)$ quantifies the field-induced coupling between the two states. The instantaneous eigenstates of $H(t)$, commonly referred to as the adiabatic states, are:

$$|w_-(t)\rangle = \cos \vartheta(t)|0\rangle - \sin \vartheta(t)|1\rangle, \quad (10)$$

$$|w_+(t)\rangle = \sin \vartheta(t)|0\rangle + \cos \vartheta(t)|1\rangle, \quad (11)$$

where the mixing angle is given by $\vartheta(t) = \frac{1}{2} \arctan \frac{\Omega(t)}{\Delta}$. The corresponding eigenvalues, representing the adiabatic energy levels, are:

$$\varepsilon_{\pm}(t) = \frac{\hbar}{2} \left(\Delta \pm \sqrt{\Omega^2(t) + \Delta^2} \right), \quad (12)$$

yielding an energy gap of $\varepsilon(t) = \varepsilon_+(t) - \varepsilon_-(t) = \hbar\sqrt{\Omega^2(t) + \Delta^2}$.

Using the eigenvectors from Eq. (11) we define the adiabatic transformation matrix R . Applying this transformation to the Hamiltonian in Eq. (9) gives the Hamiltonian in the adiabatic basis $H^a = R^\dagger H R - iR^\dagger \dot{R}$, which can be explicitly written as:

$$H^a(t) = \frac{1}{2} \begin{bmatrix} -\varepsilon(t)/2 & -i\dot{\vartheta}(t) \\ \dot{\vartheta}(t) & \varepsilon(t)/2 \end{bmatrix}. \quad (13)$$

Adiabatic evolution requires that the rate of change of the mixing angle $\vartheta(t)$ is much smaller than the energy splitting $\varepsilon(t)$, leading to the adiabatic condition [? ?]:

$$|2\dot{\vartheta}| \ll \varepsilon(t), \quad (14)$$

where the non-adiabatic coupling term is given explicitly by $|\dot{\vartheta}| = \frac{\Delta\dot{\Omega}(t)}{2[\Delta + \Omega(t)]}$.

Replacing the inequality in Eq. (14) with an equality provides a approximate boundary condition between adiabatic and non-adiabatic regions

$$\sqrt{\Omega^2(t_0) + \Delta_b^2} = \frac{\Delta_b \dot{\Omega}(t_0)}{2[\Delta^2 + \Omega^2(t_0)]}, \quad (15)$$

where t_0 denotes the time at which the non-adiabatic coupling $|\dot{\vartheta}|$, is maximal, and Δ_b marks the boundary between the adiabatic and non-adiabatic regimes. This condition allows us to extract a threshold value for the detuning range $\Delta < \Delta_b$ which marks the range of detunings where post-pulse excitation remains significant. The excitational linewidth scales proportionally to the threshold detuning $\text{FWHM} \propto \Delta_b$. Consequently, we are particularly interested in the functional dependence of Δ_b on the peak Rabi frequency Ω_0 .

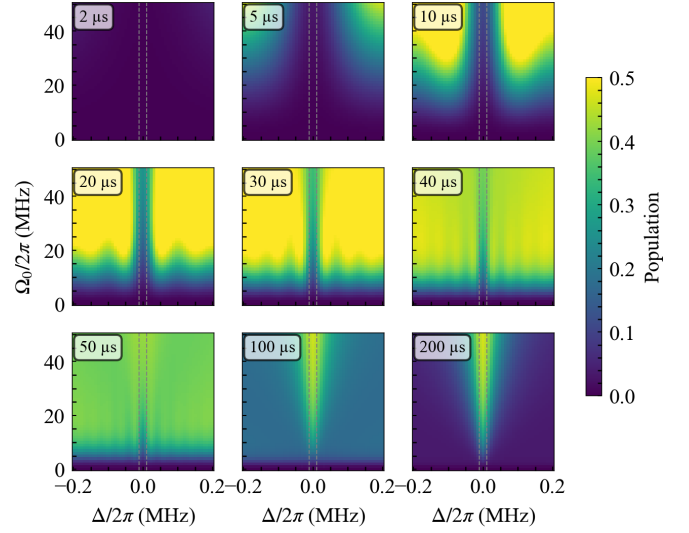


Figure 9. Numerical spectroscopic response obtained using echo-root-Lorentzian pulses for different pulse durations (indicated in each panel). As the pulse length increases, the spectral linewidth progressively narrows, approaching the decoherence-limited resolution set by the coherence time T_2 . For pulse durations comparable to or exceeding the relaxation time T_1 , population decay during the interrogation leads to an elevated off-resonant background, reducing spectral contrast despite the continued narrowing of the resonance feature.

V. PULSE LENGTH

The pulse duration imposes an additional constraint on spectroscopic resolution, as the spectral width of the drive in the frequency domain scales inversely with the pulse length. Figure 8 presents a numerical calculation of the extracted linewidth obtained using a square pulse, showing that for short pulse durations ($\leq 20 \mu s$), the measured spectroscopic linewidth is dominantly limited by Fourier broadening rather than decoherence.

In contrast, for pulse durations exceeding approximately $20 \mu s$, the echo-based pulse sequence approaches the decoherence-limited regime, yielding spectroscopy with an $\mathcal{O}(1)$ agreement with the T_2 -limited linewidth.

For longer pulse durations, an additional effect emerges: the off-resonant background signal gradually decays while the resonance feature itself remains approximately constant. This results in the formation of a characteristic narrow peak nested within the broader echo-induced dip, reflecting the onset of adiabatic state evolution in the vicinity of the resonance.

VI. HIGHER AMPLITUDES

For drive amplitudes exceeding 50 MHz , the system enters the regime of two-photon transitions. This occurs because the Rabi amplitude becomes comparable to half of the qubit anharmonicity, which corresponds to the first

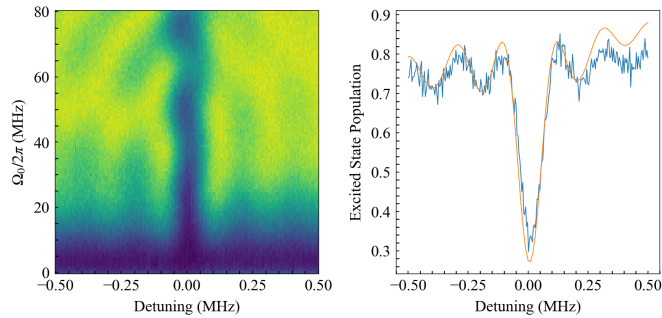


Figure 10. Two-dimensional spectroscopy map obtained with a $10\ \mu\text{s}$ echo-root-Lorentzian pulse at extended drive amplitudes (up to 80 MHz). At shorter effective pulse durations and lower Rabi rates, distortions of the resonance profile become visible due to the proximity of the first two-photon transition. The left panel shows the full detuning-amplitude map, while the right panel presents a line cut at fixed drive strength (blue) together with a smooth fit (orange), highlighting deviations from an ideal single-photon response.

two-photon transition of the system.